

energia cinética
 $L = K - U \rightarrow$ energia potencial.

vector columna $\rightarrow \mathcal{L} = \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q}$

$K(q, \dot{q}) = \frac{1}{2} \dot{q}^T M(q) \dot{q} \rightarrow K$ é função de $\dot{q}$ (ou $q(t)$)

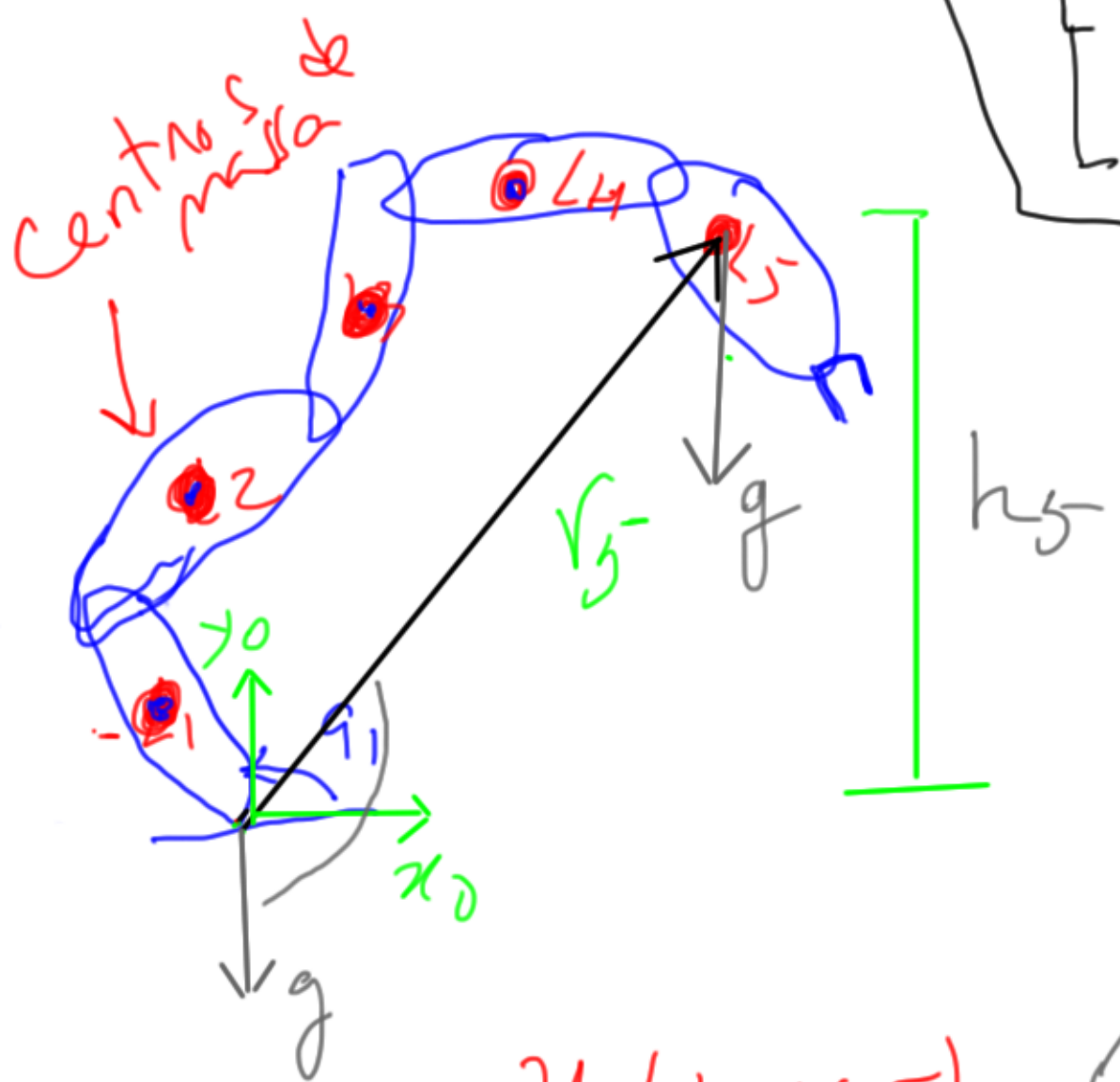
$\mathcal{L} = \frac{d}{dt} [M(q) \dot{q}] - \frac{\partial K}{\partial q} + \frac{\partial U}{\partial q}$
 $\mathcal{L} = \dot{M}(q) \dot{q} + M(q) \ddot{q} - \frac{1}{2} \dot{q}^T \frac{\partial M}{\partial q} \dot{q} + \frac{\partial U}{\partial q}$

$M(q) \ddot{q} + \underbrace{\left[\dot{M}(q) - \frac{1}{2} \dot{q}^T \frac{\partial M}{\partial q} \right]}_{C(q, \dot{q})} \dot{q} + \underbrace{\frac{\partial U}{\partial q}}_{g(q)} = \mathcal{L}$

$M(q) \ddot{q} + C(q, \dot{q}) \dot{q} + g(q) = \mathcal{L}$

$\dot{M}(q) = \frac{\partial M}{\partial q_1} \dot{q}_1 + \frac{\partial M}{\partial q_2} \dot{q}_2$

$\dot{q}^T \frac{\partial M}{\partial q} = \begin{bmatrix} \dot{q}^T \frac{\partial M}{\partial q_1} \\ \dot{q}^T \frac{\partial M}{\partial q_2} \\ \vdots \\ \dot{q}^T \frac{\partial M}{\partial q_n} \end{bmatrix}$



uma linha
 $E_p = mgh$
 $\frac{mgh}{ch \cos \theta}$

$r_5^T g = |r_5| |g| \cos(\theta)$

$a^T \cdot b = |a| \cdot |b| \cos(\alpha, b)$

$U(\text{Link } 5) = -m r_5^T g$

$U(\text{robô}) = \sum_{i=1}^n -m_i r_i^T g$

termo de gravidade $g(q) = \frac{\partial U}{\partial q} = \begin{bmatrix} \partial U / \partial q_1 \\ \vdots \\ \partial U / \partial q_n \end{bmatrix}$

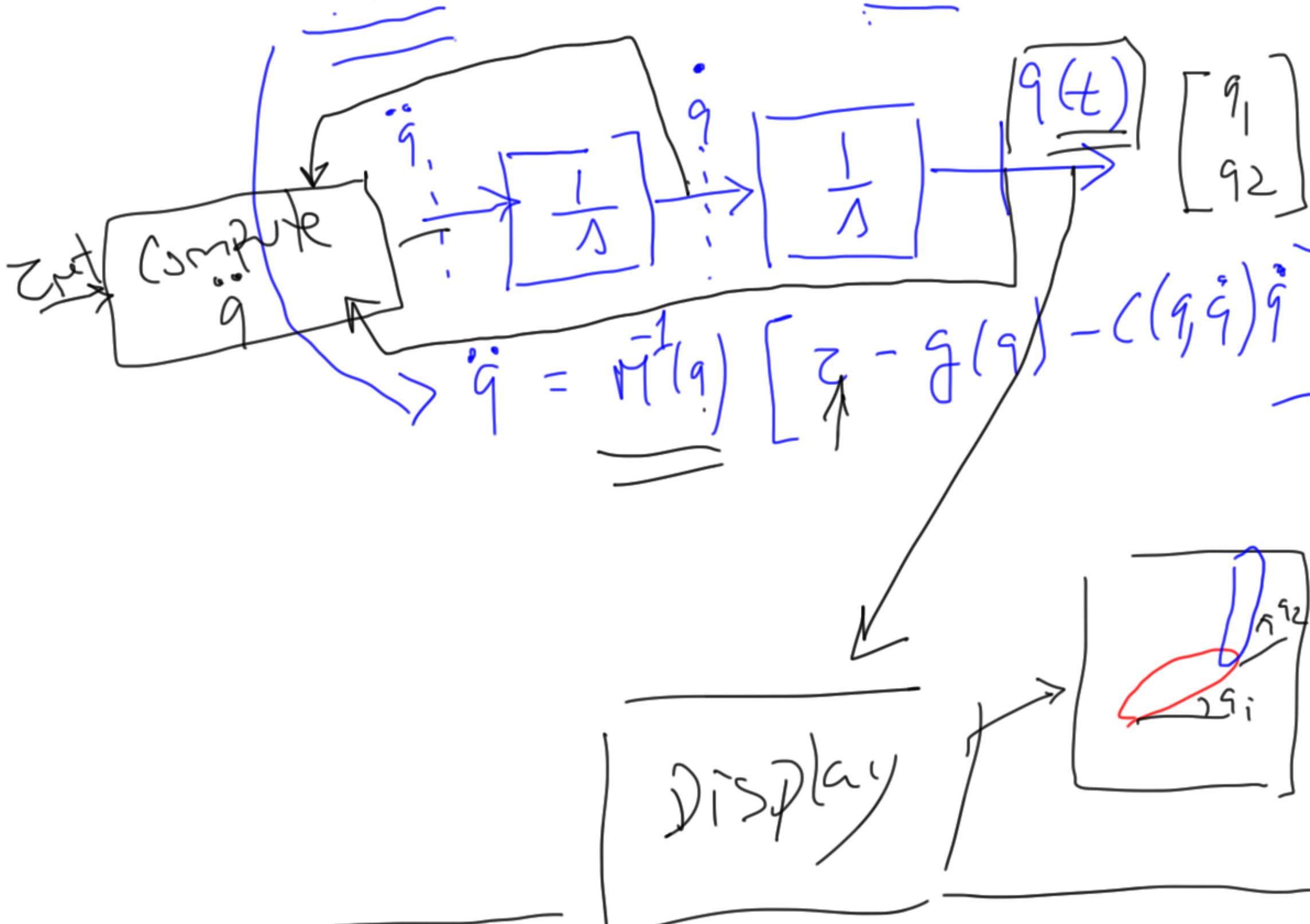
$\dot{x}_c = J_c \dot{q} \quad K = \frac{1}{2} m v^2 \quad K = \frac{1}{2} \dot{q}^T M(q) \dot{q}$

$\frac{1}{2} m_1 \dot{x}_c^T \dot{x}_c$

$K = \frac{1}{2} m_1 \dot{q}^T J_c^T J_c \dot{q}$

$M_1 = m_1 J_c^T J_c$

$M(q) \ddot{q} + C(q, \dot{q}) \dot{q} + g(q) = \mathcal{L}$



prop. da matriz de Inércia

$K = \frac{1}{2} \dot{q}^T M(q) \dot{q}$

$M(q) = M^T(q)$

$K^T = K = \frac{1}{2} \dot{q}^T M(q) \dot{q}$

matriz M é simétrica

matriz positiva definida $\mathcal{L}^T M \mathcal{L} > 0 \quad \forall \mathcal{L} \neq 0$

$\dot{X} = J \dot{q} \leftarrow \underline{X} = \underline{f(q)} = \begin{bmatrix} f_1(q) \\ \vdots \\ f_m(q) \end{bmatrix}$

$\dot{X} = \begin{bmatrix} \frac{\partial f_1}{\partial q_1} \dot{q}_1 + \frac{\partial f_1}{\partial q_2} \dot{q}_2 + \dots + \frac{\partial f_1}{\partial q_n} \dot{q}_n \\ \vdots \\ \frac{\partial f_m}{\partial q_1} \dot{q}_1 + \frac{\partial f_m}{\partial q_2} \dot{q}_2 + \dots + \frac{\partial f_m}{\partial q_n} \dot{q}_n \end{bmatrix}$

$\dot{X} = \begin{bmatrix} \frac{\partial f_1(q)}{\partial q_1} & \frac{\partial f_1(q)}{\partial q_2} & \dots & \frac{\partial f_1(q)}{\partial q_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m(q)}{\partial q_1} & \frac{\partial f_m(q)}{\partial q_2} & \dots & \frac{\partial f_m(q)}{\partial q_n} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \vdots \\ \dot{q}_n \end{bmatrix}$

Jacobiano